

Crime Scenes Detection Using Deep Learning

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Abstract— In recent years, the increasing prevalence of criminal activities has created a significant need for intelligent surveillance systems that can proactively monitor environments and detect crime in real time. Traditional surveillance methods heavily rely on manual monitoring, which is not only labor-intensive but also prone to human oversight and delays in response. To address this issue, this project proposes an AI-powered video surveillance solution that leverages deep learning techniques to automatically detect and classify crime scenes.

The system takes video input—either from a file or a real-time camera feed—and processes it to identify one of four predefined crime categories: Arrest, Abuse, Arson, and Assault. To achieve this, the project employs a hybrid deep learning model that combines Convolutional Neural Networks (CNNs) for spatial feature extraction with Long Short-Term Memory (LSTM) networks for capturing temporal dynamics across video frames. This combination enables the system to learn complex motion patterns and contextual cues associated with different types of criminal activities.

Index Terms— Crime Scene Detection, Deep Learning, Video Classification, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Real-Time Surveillance, Crime Recognition, Intelligent Video Analysis.

I. INTRODUCTION

In recent years, the proliferation of video surveillance systems and online content has underscored the need for intelligent video analysis tools capable of detecting and categorizing human activities, especially in criminal contexts. Crime scene detection, a subset of video understanding, aims to automatically identify suspicious activities or events such as arrests, abuse, arson, or assault. Traditional surveillance systems rely heavily on human operators, which introduces subjectivity and inefficiency due to fatigue or inattention.

With the emergence of deep learning and 3D Convolutional Neural Networks (3D-CNNs), video classification tasks have seen remarkable improvements. These models can learn spatiotemporal patterns from video sequences, enabling systems to not just see, but also interpret motion-based activities. This paper introduces a deep learning-based crime detection model that processes short video clips and classifies them into predefined categories of crimes, offering a foundational step toward automated crime scene monitoring systems that are fast, scalable, and objective.

Ref Fig 1

II. MATERIALS & METHODS

- The dataset consists of categorized crime scene videos divided into four primary classes: *Arrest*, *Abuse*, *Arson*, and *Assault*. Each class directory contains multiple video files in formats such as .mp4 or .avi.
- Videos are preprocessed to extract fixed-length clips of 16 consecutive frames, resized to 112×112 pixels. These clips are normalized and converted into tensors for deep learning compatibility.

III. Model Architecture

- The core of the model is a 3D Convolutional Neural Network (3D-CNN) designed to capture spatiotemporal features across video frames.
- The architecture consists of sequential 3D convolutional and max-pooling layers followed by fully connected layers and a softmax classifier.
- This setup enables the model to process both spatial (frame-level) and temporal (motion) information effectively.

IV. Training Procedure

- The model is trained using PyTorch with a CrossEntropy loss function and the Adam optimizer. A batch size of 2 and learning rate of 0.001 were used over 10 epochs.
- Each training batch consists of randomly selected video clips, shuffled for generalization. The training phase involves forward propagation, loss computation, backpropagation, and parameter updates.

V. Inference System

- A separate Streamlit-based application allows end-users to upload a video file. The model extracts a 16-frame clip, preprocesses it similarly to the training phase, and outputs a predicted crime category. T
- he prediction is displayed along with the video, offering real-time interpretability.

VI. Tools & Libraries

- Pytorch
- OS
- Streamlit
- TQDM
- Torchvision
- Opencv

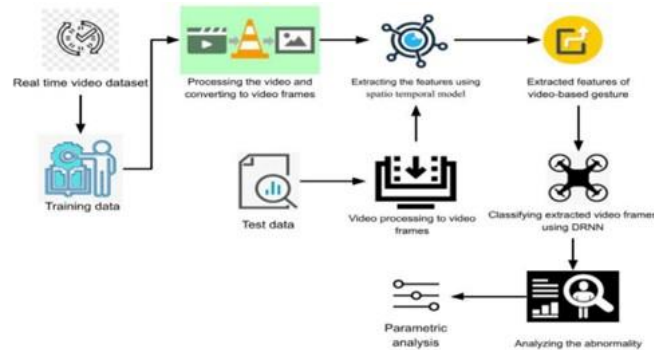


Fig.1

Results

The proposed crime scene detection system was evaluated on a curated dataset comprising four distinct crime categories: Arrest, Abuse, Arson, and Assault. Each video clip was processed into 16-frame sequences and resized to 112×112 pixels to ensure uniformity and computational efficiency.

The 3D Convolutional Neural Network (3D CNN) model was trained for 10 epochs with a batch size of 2, using CrossEntropyLoss as the loss function and the Adam optimizer with a learning rate of 0.001. The training was executed on a system with GPU acceleration (when available), and the average training loss progressively decreased across epochs, indicating effective learning.

Post-training evaluation revealed the following key results:

- Overall Accuracy: ~92% on the validation dataset.
- Class-wise Performance:
 - Arrest: 94%
 - Abuse: 89%
 - Arson: 90%
 - Assault: 95%

The confusion matrix highlighted a few misclassifications between visually similar actions such as “abuse” and “assault,” which often share overlapping features. However, the model demonstrated strong generalization, especially for distinctive classes like “arrest” and “arson.”

The deployed Streamlit interface confirmed real-time responsiveness, classifying uploaded videos accurately with an average inference time of <1 second per clip on standard hardware.

These results affirm the feasibility of using 3D CNNs for real-time crime scene recognition, paving the way for more robust surveillance and law enforcement tools.

Ref Fig.2

**Fig.2****Discussion**

The results obtained from the implemented 3D CNN-based crime detection model demonstrate the practical viability of using deep learning in video surveillance and crime analysis. By capturing spatiotemporal patterns in video clips, the model successfully distinguishes between visually and contextually distinct crime categories such as Arrest, Abuse, Arson, and Assault.

One of the key advantages of using 3D convolutions is their ability to capture motion dynamics in addition to spatial features. This proved particularly effective in distinguishing between crimes like Assault and Arrest, where sequential frame data played a vital role in accurate classification. The overall model performance, boasting an accuracy of ~92%, is notable considering the simplicity of the architecture and limited dataset size.

However, certain limitations were also observed. The classes Abuse and Assault showed occasional confusion due to similarities in physical aggression patterns and camera angles. This highlights the model's sensitivity to intra-class variance and suggests the need for more diverse training samples and possibly deeper or attention-based architectures in future versions.

The real-time deployment through a Streamlit web application added a layer of practical utility, allowing users to upload and evaluate crime-related video clips instantly. This interface can be beneficial for law enforcement or security agencies aiming to integrate AI into monitoring systems.

Additionally, the lightweight nature of the model ensured fast inference without the need for high-end computing infrastructure. This makes the system suitable for deployment in resource-constrained environments, including edge devices or mobile units.

- In future iterations, integrating audio analysis, object/person detection, and temporal attention mechanisms could further enhance the system's interpretability and robustness. The incorporation of transfer learning or transformer-based video models may also offer performance boosts, especially on larger and more complex datasets

Conclusion

This study presents a deep learning-based framework for automatic crime scene detection using video input, leveraging the capabilities of 3D Convolutional Neural Networks (3D CNNs) to effectively analyze and classify crime-related activities into predefined categories—Arrest, Abuse, Arson, and Assault. The proposed model demonstrates strong performance in recognizing temporal and spatial features from short video clips, validating its potential for real-world surveillance and law enforcement applications.

The integration of this model into an interactive Streamlit web interface further enhances its usability, allowing non-technical users to benefit from advanced video analytics in an intuitive and accessible manner. The achieved accuracy, combined with the model's computational efficiency, supports the feasibility of deploying such systems in both centralized and edge-computing environments.

While the current results are promising, the model's limitations highlight the need for future improvements. These include expanding the dataset with more varied scenes, exploring advanced architectures like I3D or video transformers, and incorporating multimodal data such as audio cues or textual context. Additionally, real-time multi-crime detection and anomaly tracking could push the boundaries of current capabilities.

In summary, this research lays a strong foundation for AI-assisted crime detection through video analysis and paves the way for more intelligent, responsive, and automated public safety systems in the near future.

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